

3	(a)	(i)	1.	$s = ut + \frac{1}{2}at^2$		
				$192 = \frac{1}{2} \times 9.81 \times t^2$	C1	
				$t = 6.3 \text{ (6.26) s}$	A1	[2]
			2.	$\max E_k (= mgh) = 0.27 \times 9.81 \times 192$	C1	
				or		
				calculation of v ($= 61.4$) and use of $E_k (= \frac{1}{2}mv^2) = \frac{1}{2} \times 0.27 \times (61.4)^2$	(C1)	
				$\max E_k = 510 \text{ (509) J}$	A1	[2]
		(ii)		velocity is proportional to time or velocity increases at a constant rate as acceleration is constant or resultant force is constant	B1	[1]
		(iii)		use of $v = at$ or $v^2 = 2as$ or $E = \frac{1}{2}mv^2$ to give $v = 61(.4) \text{ m s}^{-1}$	B1	[1]
	(b)	(i)		R increases with velocity	B1	
				resultant force is $mg - R$ or resultant force decreases	B1	
				acceleration decreases	B1	[3]
		(ii)		at $v = 40 \text{ m s}^{-1}$, $R = 0.6 \text{ (N)}$	C1	
				$0.27 \times 9.8 - 0.6 = 0.27 \times a$		
				$a = 7.6 \text{ (7.58) m s}^{-2}$	A1	[2]
		(iii)		$R = \text{weight}$ for terminal velocity	B1	
				<i>either</i> weight requires velocity to be about 80 m s^{-1} <i>or</i> at 60 m s^{-1} , R is less than weight		
				so does not reach terminal velocity	B1	[2]